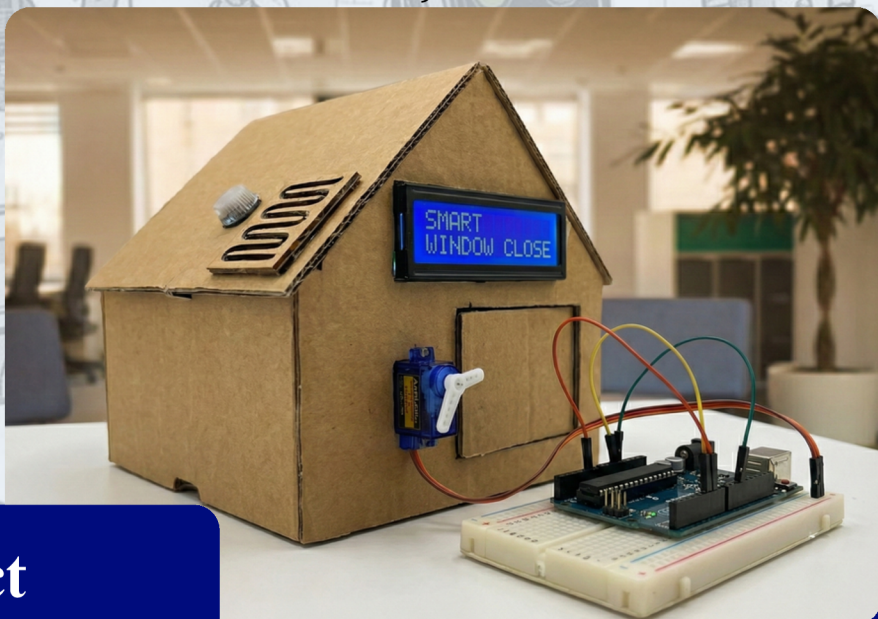


Section: 2
Group: 8
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Smart Rain Detection and Automatic Window-Closer Robot

CSE461 (Introduction to Robotics)

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Abstract

This study presents an autonomous window actuation system designed to mitigate environmental damage in residential settings. Integrating multi-sensor fusion, the system monitors real-time meteorological data-including precipitation, temperature, and humidity to execute intelligent closing maneuvers. Upon detecting rain, the control logic triggers an immediate mechanical response to secure the window, ensuring indoor protection. The design prioritizes operational safety through electromechanical limit enforcement and provides real-time status feedback via a visual interface. This research demonstrates a robust, cost-effective solution for climate-adaptive home automation and thermal management.

Introduction

- Unpredictable rainfall often causes interior water damage when manual windows are left unattended.
- This project develops an autonomous robot to detect rain and immediately close the window.
- It offers a low-cost, automated solution for smart homes and assistive living needs.

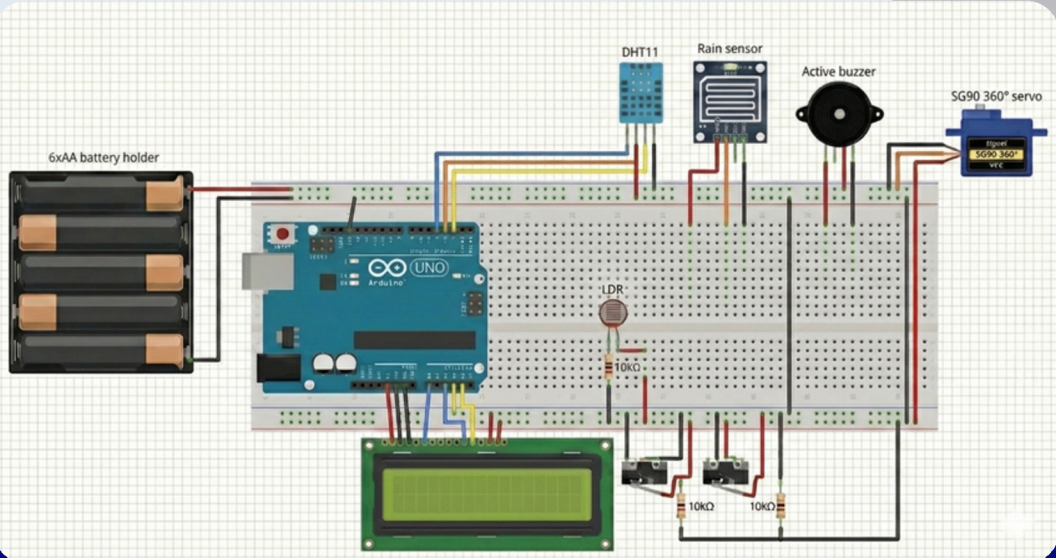
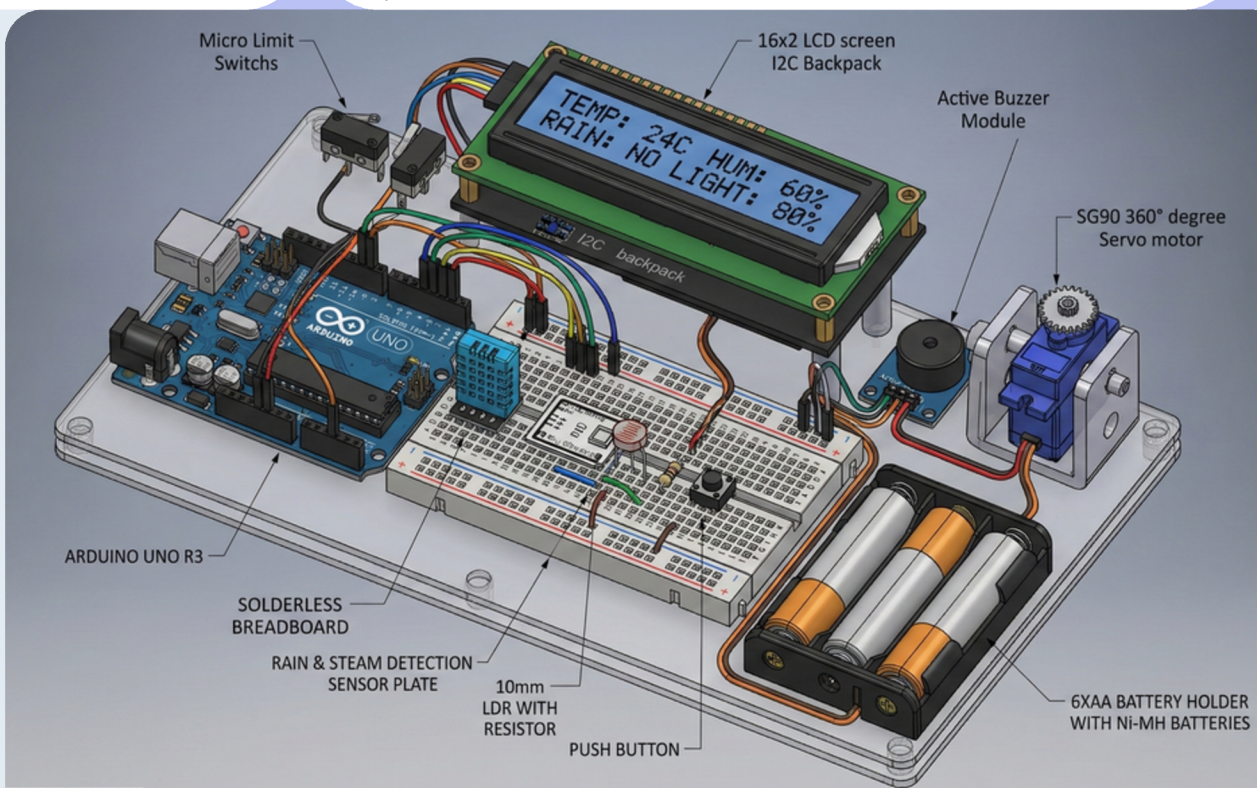
Functionalities:

- An SG90 servo drives the mechanism, utilizing limit switches to physically stop movement and prevent damage.
- Control logic triggers an automatic close response upon rain detection, supplemented by a manual override option.
- The system processes analog sensor inputs and employs the I2C protocol for efficient display communication.

Applications

- Suitable for smart homes, greenhouses, and assistive elderly care.
- Outperforms basic sensors by integrating climate data for smarter ventilation.
- A low-cost, retrofit alternative to expensive commercial systems.

- Arduino Uno R3:** Central controller processing sensor data and logic.
- Rain Sensor:** Detects precipitation to trigger immediate window closing.
- DHT11 & LDR:** Monitors real-time temperature, humidity, and light levels.
- SG90 Servo:** 360° actuator driving the window open/close mechanism.
- Limit Switches:** Safety stops to prevent mechanical over-rotation.
- 16x2 I2C LCD:** Visual display for system status and weather data.
- Active Buzzer:** Provides audio alerts during rain detection events.



Conclusion

- The successful implementation of multi-sensor fusion and safety mechanisms demonstrates a reliable, autonomous solution for residential rain protection.
- Future integration of the wind speed sensor & high-torque linear actuators will turn it into an industrial-grade window.

References

- [1] Arduino AG, "Arduino Uno Rev3 Technical Reference," Official Hardware Documentation, 2024. [Online].
- [2] Aosong Electronics, "DHT11 Temperature and Humidity Sensor Module," Product Datasheet, 2020.